Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[22ESEE11]

FIRST SEMESTER B.Tech EXAMINATION, FEBRUARY - 2023

BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

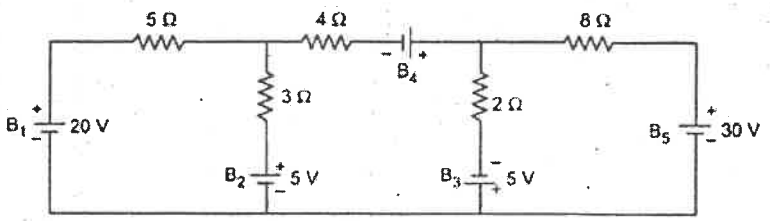
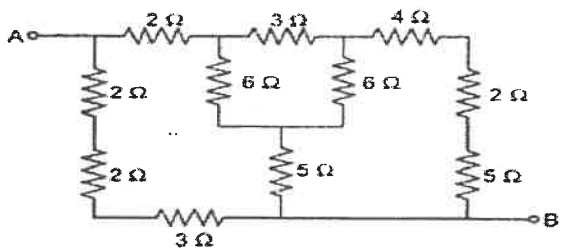
Instructions: **i.** Missing data may be suitably assumed.

ii. Answer All Questions

PART - A

5 X 2=10

1.	**a.**	State Ohm's law and mention its equation.	L-1	CO1	**02**
	b.	Discuss form factor and peak factor.	L-2	CO2	**02**
	c.	List the characteristics of MCB.	L-1	CO3	**02**
	d.	Describe wire-wound resistors.	L-2	CO4	**02**
	e.	Sketch the V-I characteristics of PN junction diode.	L-2	CO6	**02**
		PART - B			**6 X 5=30**
2.	**a.**	A 100-watt, 250 V lamp is connected in series with a 100-watt, 200 V lamp across 250 V supply. Calculate **i.** circuit current and **ii.** voltage across each lamp. Assume the lamp resistances to remain unaltered.	L-3	CO1	**05**
		OR			
	b.	State and explain Kirchhoff's laws.	L-2	CO1	**05**
3.	**a.**	A coil when connected across a 100 V d.c. supply dissipates 500 W of power. When connected across a 100 V a.c. supply of frequency 50 Hz, it dissipates 200 W. Calculate the values of resistance and inductance of the coil.	L-3	CO2	**05**
		OR			
	b.	A choke coil takes a current of 2A lagging 60° behind the applied voltage of 200 V at 50Hz. **i.** Calculate impedance, resistance and inductance of the coil. **ii.** Also find the power consumed when the coil is connected across 100 V, 25 Hz supply.	L-3	CO2	**05**
4.	**a.**	Describe wire earthing and plate earthing.	L-2	CO3	**05**
		OR			
	b.	Classify the various types of wiring and explain any two types of wiring in detail.	L-2	CO3	**05**

5.	**a.**	Discuss the important characteristics of batteries.	L-2	CO4	**05**
		OR			
	b.	Compare primary and secondary batteries.	L-2	CO4	**05**
6.	**a.**	A 9V power supply is to be tested with a 39Ω load resistor. If the resistor has a tolerance of 10%. Determine **i.** the nominal current taken from supply, **ii.** the maximum and minimum values of supply current at either end of tolerance range for the resistor	L-3	CO5	**05**
		OR			
	b.	Explain Capacitance along with its equation and classify its types according to structure.	L-2	CO5	**05**
7.	**a.**	Illustrate the circuit of bridge rectifier and explain its operations with waveforms.	L-3	CO6	**05**
		OR			
	b.	Compare half wave and full wave rectifier.	L-2	CO6	**05**
		PART - C			**6 X 10=60**
8.	**a.**	Determine the current supplied by each battery as shown in circuit by using kirchoff's law.  	L-3	CO1	**10**
		OR			
	b.	Calculate the effective resistance between the points A and B in the given circuit.  	L-3	CO1	**10**
9.	**a.**	Illustrate and explain the RLC series circuit diagram and phasor diagram with different cases .	L-3	CO2	**10**
		OR			

	b.	A voltage v(t) = 100 sin 314 t is applied to a series circuit consisting of 10Ω resistance, 0·0318 H inductance and a capacitor of 63.6 μF. Find **i.** expression for I(t) **ii.** phase angle between voltage and current **iii.** power factor **iv.** active power consumed **v.** peak value of pulsating power.	L-3	CO2	**10**
10.	**a.**	Sketch the circuit of two way and three-way switch connections and explain its operation with the truth table.	L-3	CO3	**10**
		OR			
	b.	Classify the various types of cables and explain any three of its types with neat diagram.	L-2	CO3	**10**
11.	**a.**	Show the average power consumed by a pure inductance is zero. Explain with suitable equations.	L-2	CO4	**10**
		OR			
	b.	Calculate the cost of average monthly consumption for the month of June with the following data. Cost of 1unit is Rs. 5	L-2	CO4	**10**

Name of the appliance	Power rating in Watts	No. of appliances	Usage (hrs/day)
Tube light	40	8	6
Fan	80	4	4
Laptop	40	2	6
CFL	10	4	4

12.	**a.**	Sketch the block diagram of DC power supply and explain the various components associated with it.	L-2	CO5	**10**
		OR			
	b.	With a neat block diagram of electronic instrumentation system and explain its operation.	L-2	CO5	**10**
13.	**a.**	Illustrate the principle of operation of PN junction diode along with its applications.	L-2	CO6	**10**
		OR			
	b.	Explain the working of simple Zener diode regulator along with its characteristics. List the applications of Zener diode.	L-2	CO6	**10**